

their physical proximity. Quantum superposition is the phenomenon by which a quantum system can be in a combination of several separate quantum states at the same time. Notice that the above two definitions are not rigorous enough to capture the full intricacies of these quantum mechanical phenomena. A quantum computer is a device that uses these quantum mechanical phenomena to perform computations. In quantum computers we have qubits (quantum bits) for data storage in place of bits used in the classical computers of today.

The quantum mechanical phenomenon described earlier called superposition makes qubits far more powerful than classical bits, which in turn makes quantum computers far superior to classical computers. What makes a qubit more powerful than a bit in a classical computer? In a traditional bit, we can store either a 0 or a 1. The corresponding basis values stored in qubits are represented by $|0\rangle$ and $|1\rangle$. The power of a qubit comes from the fact that it can store a combination of the basis states of $|0\rangle$ and $|1\rangle$, in addition to the basis states themselves, due to quantum superposition. Notice that a classical bit will be in either of the two states of 0 and 1, whereas the qubit can also be in a superimposed state of $|0\rangle$ and $|1\rangle$ and not just any one of them. The difficulty in manufacturing quantum computers arises from the fact that in order to make qubits, we need objects that can attain a state of quantum superposition between two states.

There are different ways to create a system capable of exhibiting quantum superposition. Some of the techniques involve suspending ions in vacuum by using an electromagnetic field, positioning single atoms in crystals, using photons in optical fibres, etc. But all these techniques are difficult to implement, especially when the number of qubits in a system increases. This makes the creation of quantum computers extremely difficult. Another

term often associated with quantum computing is quantum supremacy. It is a term defined by John Preskill, a theoretical physicist, and it denotes the ability of a quantum computer to get a solution to a problem for which no solution is feasible by a classical computer. For example, for a classical computer it is extremely difficult to find the factors of a large semiprime (a number that is a multiple of two prime numbers). If the number is sufficiently large, even the fastest supercomputer in the world will take thousands of years to find the factors of semiprimes (this is what makes public key cryptography effective). A quantum algorithm called Shor's algorithm can solve this problem of integer factorisation in polynomial time, making it a feasible solution.

So a quantum computer can solve the problem of integer factorisation in minutes or even seconds, instead of the thousands of years required by even the fastest classical computer of today. Quantum cryptography is yet another term often associated with quantum computing. It involves the use of quantum mechanical properties for cryptographic purposes. Strictly speaking, quantum cryptography has an existence that is independent of quantum computing. Notice that post-quantum cryptography (the study of cryptographic techniques that will be able to withstand the attack of quantum computers) is distinct from quantum cryptography. For example, in a world where quantum computers are common, public key cryptography as we know today will not be secure enough to be used in sensitive fields like banking, commerce, etc.

The power of quantum computers

With the necessary vocabulary about quantum computing built, let us now discuss the potential power of quantum computers, if and when they are ready and deployed. In what ways are they superior to classical computers? Consider a classical computer with N bits of storage. It can store a single number

between 0 and 2^N-1 in binary. Now consider a quantum computer with N qubits of storage. Any number between 0 and 2^N-1 can be stored in this qubit storage but with a certain probability for the presence of each number.

At this stage, I would like to point out a misconception about the way qubits work. People often believe that we can store all the numbers between 0 and 2^N-1 in N qubits — all at the same time. This is not the case. What we can say is that multiple readings of the N qubits in a quantum computer may yield different numbers, and the power of quantum computing comes from this fact. To further elaborate the additional power we can extract from a quantum computer, let us try to understand complexity classes.

A computational problem is said to be in complexity class P , if it can be solved by a classical computer in polynomial time. Let us say that if a computational problem is in class P , then we can solve the problem in a reasonable amount of time even if the problem input is sufficiently large, by using a classical computer. Examples for problems in class P include addition, subtraction, multiplication and division of numbers, sorting a list of numbers, searching for a number in a list, checking whether a graph contains an Eulerian cycle, etc. In general, we can say that problems in class P have a feasible solution. There are many other complexity classes also, all of which in some sense tell us how difficult it is to obtain a solution to a computational problem. For example, there are no known polynomial time (feasible) solutions to problems in class NP . Examples for problems in class NP include the Hamiltonian cycle problem, travelling salesman problem, clique problem, subset sum problem, etc. It is not known whether polynomial time solutions to these problems exist. In fact, the $P=NP$ problem is one of the most important open problems in computer science. You could win a million dollars by solving it!